DOXORUBICIN

Cardiotoxic — lifetime max dose

Vesicant

CHEMICAL NAME (8S,10S)-10-[(3-amino-2,3,6-trideoxy-α-L-lyxo-hexopyranosyl)oxy]-6,8,11-trihydroxy-8-(hydroxyacetyl)-1-methoxy-7,8,9,10-tetrahydrotetracene-5,12-dione (hydrochloride)

CLASS Anthracycline cytotoxic antibiotic

TRADE NAMES Adriamycin · Caelyx / Doxil (liposomal) · Doxorubicin injection

THERAPEUTIC USES / INDICATIONS

- Breast cancer (AC regimens).
- Lymphomas (CHOP, ABVD), acute leukaemias.
- Sarcomas, ovarian, bladder, gastric cancers; childhood tumours.
- Liposomal: ovarian cancer, Kaposi's sarcoma, myeloma.

ADMINISTRATION

- **IV** via free-flowing drip or central line — vesicant (extravasation → severe necrosis; antidote dexrazoxane).
- Never IM/SC; not liposomal and conventional interchangeable.
- Cytotoxic precautions; red urine for 1–2 days.

DOSAGES

- **Typical:** 60–75 mg/m² every 3 weeks (AC: 60 mg/m²).
- **Cumulative lifetime max:** 450–550 mg/m² (lower with chest radiotherapy, trastuzumab, cardiac disease).
- Reduce in hepatic impairment (bilirubin).

PHARMACOKINETICS

- Extensively distributed; metabolised in liver to active doxorubicinol.
- Excreted in bile (and urine — red colour). **t½:** 20–48 h.

PHARMACODYNAMICS

- Intercalates into **DNA**, inhibits **topoisomerase II**, generates **free radicals** → DNA damage and cell death (also cardiomyocyte injury).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Alopecia (total), nausea, vomiting.
- Mucositis.
- Red/orange urine (harmless).

ADVERSE EFFECTS (SERIOUS)

- Cardiotoxicity: acute arrhythmias; dose-dependent cardiomyopathy and heart failure (may appear years later).
- Extravasation tissue necrosis.
- Myelosuppression.
- Secondary leukaemia; radiation recall; hand–foot syndrome (liposomal).

CONTRAINDICATIONS & CAUTIONS

- Severe myelosuppression; severe hepatic impairment.
- Significant cardiac impairment (reduced LVEF), recent MI, severe arrhythmias.
- Previous maximum cumulative anthracycline dose.
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **Trastuzumab, cyclophosphamide, radiotherapy:** ↑ cardiotoxicity.
- **Paclitaxel:** ↑ doxorubicin levels.
- **Live vaccines:** contraindicated.

NURSING MANAGEMENT

- Baseline and periodic echocardiogram/MUGA (LVEF); track cumulative dose.
- Check cannula patency; observe site throughout; stop immediately if pain/swelling; extravasation protocol.
- FBC before each cycle; antiemetics.
- Scalp cooling may reduce hair loss.

PATIENT EDUCATION

- Urine may be red for 1–2 days — harmless.
- Report pain, burning or swelling at the drip site immediately.
- Report breathlessness, swelling, palpitations (even years later).
- Report fever ≥38 °C; expect hair loss.

KEY POINTS

- Anthracycline; topo II inhibitor, DNA intercalator.
- Vesicant; extravasation protocol.
- Cardiotoxic — lifetime max 450–550 mg/m².
- Red urine; myelosuppression.

SOURCES CHECKED **[892]** Doxorubicin 2 mg/ml Concentrate for solution for infusion (UK SmPC) **[893]** Doxorubicin Hydrochloride, IV (US label)

Full references p. 432–457